

Evert Nasedkin, PhD

Curriculum Vitae

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Nationality - Canadian

Through my training as an astronomer I have gained extensive experience in scientific data analysis, modelling of physical processes, open-source software development, and in the communication of complex results to a variety of audiences. I am eager to leverage these skills in new areas that are – literally – more down to earth.

Experience

- 2024-Now **Postdoctoral Fellow**, *School of Physics*, Trinity College Dublin, Dublin, IE.
- Implemented time-resolved retrieval framework for brown dwarf atmospheres.
 - Co-authored multiple high-impact publications, presented findings at premier international conferences, and organised media releases for key results.
 - Led successful observing program using the James Webb Space Telescope.
 - Member of the scientific organising committee for a session at the European Astronomical Society meeting.
 - Trained bachelor's student in the use of python for scientific computing.
- 2020-2024 **PhD Candidate**, *Max Planck Institute for Astronomy*, Heidelberg, DE.
- Developed novel atmospheric retrieval framework for exoplanet atmospheres as part of open-source `petitRADTRANS` package.
 - Obtained and analysed large high-contrast imaging datasets using a range of python-based image processing tools.
 - Published 3 first-author papers, presented results at 11 international conferences.
 - Organised multiple workshops and seminars with an international audience.
 - Served as PhD student representative, led efforts to improve student onboarding processes and access to resources.
- 09-12 2019 **Semester Project**, *Institute for Astronomy*, ETH Zürich, Zürich, CH.
- Processed and visualised 3D hydrodynamic simulations of planet formation.
- 05-08 2017 **Research Assistant**, *Institute for Astronomy*, ETH Zürich, Zürich, CH.
- Designed and performed mechanical tests at cryogenic conditions for astronomical instrumentation.
- 08-12 2016 **Research Assistant**, *nEXO Collaboration*, McGill University, Montreal, QC.
- Simulated and assembled electroluminescent source for the nEXO neutrino experiment.
- 2015-2016 **Research Assistant**, *DEAP-3600 Dark Matter Search*, SNOLAB, Sudbury, ON.
- Implemented and automated analyses for characterising detector behaviour.
- 2014-2016 **Aerodynamics Team Member**, *FSAE Student Design Team*, Waterloo, ON.
- Performed computational fluid dynamics experiments; assisted with manufacturing of aerodynamic systems.

Education

- 2020-2024 **Dr. Rer. Nat., Astronomy**, University of Heidelberg, Heidelberg, DE,
Max Planck Institute for Astronomy, Heidelberg, DE.
Dissertation: "Atmospheric Characterisation of Directly Imaged Exoplanets".
- 2018-2020 **MSc, Physics**, ETH Zürich, Zürich, CH.
Thesis: "Sub-Stellar Atmospheres in the Mid-Infrared".
- 2013-2018 **BSc, Honours Co-operative Physics**, University of Waterloo, Waterloo, ON.

Selected Publications

- **Nasedkin** et al. "The JWST Weather Report: retrieving temperature variations, auroral heating, and static cloud coverage on SIMP-0136", submitted A&A, (2025).
- Hoch et al. "Silicate clouds and a circumplanetary disk in the YSES-1 exoplanet system", Accepted Nature, (2025).
- **Nasedkin** et al. "Four-of-a-kind? Comprehensive atmospheric characterisation of the HR 8799 planets with VLT/GRAVITY", A&A, 687, A298 (2024).
- **Nasedkin**, Mollière, and Blain "Atmospheric retrievals with petitRADTRANS", JOSS 9, 96, 5875 (2024).
- **Nasedkin**, et al. "Impacts of high-contrast image processing on atmospheric retrievals", A&A, 678, A41 (2023).
- Vasist et al. "Neural posterior estimation for exoplanetary atmospheric retrieval", A&A, 672, A147 (2023).
- Patapis, **Nasedkin** et al. "Direct emission spectroscopy of exoplanets with the medium resolution imaging spectrometer on board JWST MIRI. I. Molecular mapping and sensitivity to instrumental effects", A&A, 658, A72 (2022).

Computing Experience

Projects

petitRADTRANS

Codeveloper of widely used pRT package (>300 citations). Wrote module to perform atmospheric retrievals on directly imaged and transiting exoplanets.

<https://gitlab.com/mauricemolli/petitRADTRANS>

Data reduction for IFUs

Developed pipelines to streamline the post-processing of high-contrast hyper-spectral data using state-of-the-art algorithms.

PACO

Contributed a python implementation of the PATCH COVariance (PACO) PSF subtraction algorithm to the open-source VIP-HCI package.

Computing Resources

HPC Extensive experience using the SLURM based VERA and VIPER clusters (MPCDF).

Tools Use of Linux, Mac, and Windows, together with bash scripting and command line interface.

git Version control, automated testing, and package release via PyPI.

Programming Languages

Python Expert – Open source design and development, extensive use of scientific and data analysis packages (numpy, scipy, JAX, pytorch, scikit-learn, pandas).

Fortran Intermediate – Contributed to pRT radiative transfer backend.

C++ Intermediate – Automated analysis for DEAP-3600 detector characterisation.

Languages

English: Native speaker. **Spanish:** A2. **German:** A1.

Personal Interests

Photography, Cycling, Triathlon, Music, Hiking.